TED (21)3341
(Revision – 2021)

2109230383

Reg. No……
Signature…….

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/ MANAGEMENT/COMMERCIAL PRACTICE, NOVEMBER – 2025

DISCRETE MATHEMATICS

[Maximum Marks: **75**] [Time: **3** Hours]

PART-A

I. Answer '*all*' the following questions in one word or one sentence. Each question carries *'one'* mark.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1.	Interpret the logically equivalence to the statement "I pass only if you pass"?	M1.01	A
2.	Write the dual for $p \leftrightarrow q$.	M1.03	A
3.	Give example for a finite set.	M2.01	U
4.	If $(x^2-1, y+3) = (3,4)$, find the values of x and y.	M2.01	U
5.	Define injective function.	M2.07	R
6.	State Pigeon hole Principle.	M3.03	R
7.	If a student can choose a project either 20 from mathematics or 35 from CS or 15 from engineering. In how many ways a student can choose a project.	M3.02	A
8.	Define semi group with example.	M4.02	R
9.	Define a lattice.	M4.03	R

PART-B

II. Answer any '*eight*' questions from the following. Each question carries *'three'* marks.

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1.	Show that the implication $\neg(p \rightarrow q) \rightarrow \neg q$ is tautology.	M1.03	A
2.	Without constructing a truth table, prove that $(P \wedge Q) \Rightarrow P \rightarrow Q$.	M1.05	A
3.	If A = {1, 2, 3, 4}, give an example of a relation on A which is reflexive and transitive, but not symmetric.	M2.06	U
4.	Find all subset of the set {1, 2, 3}.	M2.01	U
5.	If f : $R \rightarrow R$ and g: $R \rightarrow R$ are given by f(x) = sin x and g (x)=$5x^2$, then find g ∘ f(x)	M2.07	A
6.	Let set X = {1, 2, 3, 4} and relation R = {(x, y) |x > y}. Draw the graph of R and also give its matrix.	M2.05	U
7.	In how many ways can the letters of the word 'MATHEMATICS' be arranged such that vowels must always come together?	M3.03	A
8.	Find the 10th term in the expansion of $\left(x^2 + \frac{1}{x^2}\right)^{20}$.	M3.04	U
9.	Prove that (Z, +) is a monoid.	M4.02	U
10.	If H and K are subgroups of a group G, prove that H∩K is also a subgroup of G.	M4.02	U

PART-C

Answer '*all*' questions from the following. Each question carries *'seven'* marks.

(6 x 7 = 42 Marks)

		Module Outcome	Cognitive level
III.	Construct the truth table for $p \rightarrow (q \rightarrow s) \wedge (\neg r \vee p) \wedge q$.	M1.02	A
	OR		
IV.	Write the converse, inverse and contrapositive of the implication "If $-1 < 3$ and $3 + 7 = 10$, then $sin\left(\frac{3\pi}{2}\right) = -1$". Also determine its truth value as well as the truth value of its corresponding converse, inverse and contrapositive.	M1.04	A
V.	Establish the validity of the following argument by the method of proof. $(\neg p \vee q) \rightarrow r$ $r \rightarrow (s \vee t)$ $\neg s \wedge \neg u$ $\neg u \rightarrow \neg t$ $\therefore p$	M1.05	A
	OR		
VI.	List any seven rule of Inference with related logical implication and name of the rule.	M1.05	U
VII.	a) Among a group of students, 49 study Physics, 37 study English and 21 study Biology. If 9 of these students study Physics and English, 5 study English and Biology, 4 study Physics and Biology and 3 study Physics, English and Biology, find the number of students in the group.	M2.03	A
	b) Find the inverse for f(x) = 6x + 3. Verify your inverse by computing of the composition.	M2.07	U
	OR		
VIII.	Explain different types of relations in sets with examples.	M2.04	U
IX.	An urn contains 15 balls, 8 of which are red and 7 are black. In how many ways can 5 balls be chosen so that (i) all the five are red (ii) 2 are red and 3 are black (iii) At least 3 are red.	M3.02	A
	OR		
X.	Solve the recurrence relations $a_{r+2} - 3a_{r+1} + 2a_r = 0$ by the method of generating function with the initial condition $a_0 = 2$ and $a_1 = 3$.	M3.06	A
XI.	Solve the recurrence relation $a_{n+z} = a_{n+1} + a_n$, $n \geq 0$, $a_0 = 0$, $a_1 = 1$.	M3.05	A
	OR		
XII.	State Binomial Theorem. Identify the middle term in the expansion of $(x + 2y)^7$.	M3.04	U

XIII.	Let Z be the set of integers. Define * in Z by $a * b = a + b + 2$ for all $a, b \in Z$. Verify whether (Z, *) is a group.	M4.02	U
	OR		
XIV.	a. Define homomorphic functions. Consider (Z, +) and (Z_4, +4) defining g: $Z \rightarrow Z_4$ by $g(a) = a(mod\ 4)$. Show that g is a homomorphism.	M4.01	U
	b. Describe different types of lattices.	M4.03	R

****